

Actividad de clase 2.

Servicio DNS en Ubuntu



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Parte Práctica

Paso 0: Creación de dos máquinas virtuales nuevas Ubuntu Server 22

Resultado de este Paso: Dos pantallazos, uno por cada servidor donde se haga la consulta a las direcciones IP del mismo con el nombre de la máquina virtual visible.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d3:eb:01 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.11.144/24 metric 100 brd 192.168.11.255 scope global dynamic ens33
        valid_lft 1798sec preferred_lft 1798sec
    inet6 fe80::20c:29ff:fed3:eb01/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d3:eb:0b brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 172.16.30.11/24 brd 172.16.30.255 scope global ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fed3:eb0b/64 scope link
        valid_lft forever preferred_lft forever
administrador@sd-ns1:~$ hostname
sd-ns1.2asir.com
```

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:~$ sudo netplan apply
administrador@sd-ns2:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:8e:65:2e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.11.145/24 metric 100 brd 192.168.11.255 scope global dynamic ens33
        valid_lft 1799sec preferred_lft 1799sec
    inet6 fe80::20c:29ff:fe8e:652e/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:8e:65:38 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 172.16.30.12/24 brd 172.16.30.255 scope global ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe8e:6538/64 scope link
        valid_lft forever preferred_lft forever
administrador@sd-ns2:~$ hostname
sd-ns2.2asir.com
```

Paso 1: Instalación de BIND

Resultado de este paso:

Dos pantallazos, uno por cada servidor donde se vea el estado del servicio BIND. No debe tener ningún error.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:~$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
  Active: active (running) since Tue 2024-11-19 18:34:07 UTC; 40s ago
  Docs: man:named(8)
  Process: 2261 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
  Main PID: 2262 (named)
  Tasks: 14 (limit: 4520)
  Memory: 8.8M
  CPU: 60ms
  CGroup: /system.slice/named.service
          └─2262 /usr/sbin/named -u bind

nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001:505:9000::1
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001:505:9000::1
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001:505:9000::1
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving 'com/DNSKEY/IN': 2001:505:9000::1
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving 'com/DNSKEY/IN': 2001:505:9000::1
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns2.2asir.com/A' after disabling
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns2.2asir.com/AAAA' after disabling
nov 19 18:34:18 sd-ns1.2asir.com named[2262]: network unreachable resolving 'archive.ubuntu.com/NS/IN': 2001:505:9000::1
nov 19 18:34:18 sd-ns1.2asir.com named[2262]: network unreachable resolving 'archive.ubuntu.com/NS/IN': 2001:505:9000::1
lines 1-22/22 (END)
```

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:~$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
  Active: active (running) since Tue 2024-11-19 18:34:20 UTC; 47s ago
  Docs: man:named(8)
  Process: 3099 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
  Main PID: 3100 (named)
  Tasks: 14 (limit: 4520)
  Memory: 8.2M
  CPU: 50ms
  CGroup: /system.slice/named.service
          └─3100 /usr/sbin/named -u bind

nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './NS/IN': 2001:500:2f::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './NS/IN': 2001:500:1::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './NS/IN': 2001:500:a8::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './DNSKEY/IN': 2801:1b8::1
nov 19 18:34:20 sd-ns2.2asir.com named[3100]: network unreachable resolving './NS/IN': 2801:1b8:10::1
nov 19 18:34:21 sd-ns2.2asir.com named[3100]: managed-keys-zone: Initializing automatic trust anchors
nov 19 18:34:21 sd-ns2.2asir.com named[3100]: resolver priming query complete: success
lines 1-22/22 (END)
```

Paso 2: Crear la zona de búsqueda directa

Resultado de este Paso:

- 1 Pantallazo con el contenido del fichero de declaración de la zona de búsqueda directa

```
SD_NS1 x SD_NS2 x
1 //
2 // Do any local configuration here
3 //
4
5 // Consider adding the 1918 zones here, if they are not used in your
6 // organization
7 //include "/etc/bind/zones.rfc1918";
8
9 zone "2asir.com" {
10     type master;
11     file "/etc/bind/zones/db.2asir.com";
12 };
13
14
```

- 1 Pantallazo con el fichero de zona de búsqueda directa modificado con los comentarios.

```
SD_NS1 x SD_NS2 x
1 ;Establecemos el tiempo que conservará en caché el registro antes de refrescarse
2 $TTL 604800
3 ;Configuramos el nombre de dominio de la zona, que se asigna si el valor al que apuntamos no
4 ;tiene uno vinculado ya
5 $ORIGIN 2asir.com.
6 ;Start Of Authority, que es el registro que define las características clave de la zona
7 ;Especificamos nombre de la máquina y el del administrador en esa máquina
8 @ IN SOA sd-ns1.2asir.com. admin.2asir.com. (
9     1 ; Serial - número de serie de la zona
10     604800 ; Refresh - tiempo de refresco para los servidores esclavos
11     86400 ; Retry - tiempo de reintento de contacto del esclavo al DNS
12     ;maestro
13     2419200 ; Expire - tiempo de expiración de los servidores esclavos
14     ;tiempo tras el cual el esclavo deja de contactar con el maestro
15     604800 ) ; Negative Cache TTL - tiempo que se almacena una consulta
16     ;no resuelta en caché, transcurrido se reintenta
17 ;Definimos nuestros servidores DNS autoritativos que resolverán consultas
18 @ IN NS sd-ns1
19 @ IN NS sd-ns2
20 ;El @ hace referencia al nombre del dominio (lo establecido en origen)
21 ;Establecemos la resolución de dichos DNS, para que si se busca por nombre se resuelva
22 ;a IP.
23 sd-ns1 IN A 172.16.30.11
24 sd-ns2 IN A 172.16.30.12
25 ;configuramos más RR A
26 ;en este caso todas las consultas con dominio 2asir.com irán a la 172.16.30.11
27 @ IN A 172.16.30.11
28 www IN A 172.16.30.20
29 www2 IN A 172.16.30.21
30 intranet IN A 172.16.30.22
31 ;el apodo www3 hace referencia a intranet, si se busca www3 se redirige al valor de intranet
32 www3 IN CNAME intranet
33 ;puntero a un mail
34 @ IN MX 10 mail
35 mail IN A 172.16.30.30
```

- 1 Pantallazo con el resultado de los dos chequeos de la zona creada.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:/etc/bind$ named-checkconf
administrador@sd-ns1:/etc/bind$ named-checkzone "2asir.com" "/etc/bind/zones/db.2asir.com"
zone 2asir.com/IN: loaded serial 1
OK
administrador@sd-ns1:/etc/bind$
```

- 1 Pantallazo con la verificación del estado del servicio BIND en -ns1.2asir.com. No debe tener ningún error.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:/etc/bind$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2024-11-19 18:34:07 UTC; 28min ago
     Docs: man:named(8)
   Process: 2261 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
  Main PID: 2262 (named)
    Tasks: 14 (limit: 4520)
   Memory: 8.9M
      CPU: 125ms
   CGroup: /system.slice/named.service
           └─2262 /usr/sbin/named -u bind

nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001:5
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving 'com/DNSKEY/IN': 2001:5
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: network unreachable resolving 'com/DNSKEY/IN': 2001:5
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns2.2asir.com/A' after disablin
nov 19 18:34:09 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns2.2asir.com/AAAA' after disab
nov 19 18:34:18 sd-ns1.2asir.com named[2262]: network unreachable resolving 'archive.ubuntu.com/NS/IN': 2001:5
nov 19 18:34:18 sd-ns1.2asir.com named[2262]: network unreachable resolving 'archive.ubuntu.com/NS/IN': 2001:5
nov 19 19:00:58 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns1.2asir.com/A' after disablin
nov 19 19:00:58 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns1.2asir.com/AAAA' after disab
lines 1-22/22 (END)
```

Paso 3: Crear la zona de búsqueda inversa

Resultado de este Paso:

- 1 Pantallazo con el contenido del fichero de declaración de la zona de búsqueda inversa

```
SD_NS1 x SD_NS2 x
1 //
2 // Do any local configuration here
3 //
4
5 // Consider adding the 1918 zones here, if they are not used in your
6 // organization
7 //include "/etc/bind/zones.rfc1918";
8
9 zone "2asir.com" {
10     type master;
11     file "/etc/bind/zones/db.2asir.com";
12 };
13
14 zone "30.16.172.in-addr.arpa" {
15     type master;
16     file "/etc/bind/zones/db.30.16.172.in-addr.arpa";
17 };
18
```

- 1 Pantallazo con el fichero de zona de búsqueda inversa modificado con los comentarios.

```
SD_NS1 x SD_NS2 x
1 ;Establecemos el tiempo que conservará en caché el registro antes de refrescarse
2 $TTL 604800
3 ;Configuramos el nombre de dominio de la zona, que se asigna si el valor al que apuntamos no
4 ;tiene uno vinculado ya
5 $ORIGIN 30.16.172.in-addr.arpa.
6 ;Start Of Authority, que es el registro que define las características clave de la zona
7 ;Especificamos nombre de la maquina y el del administrador en esa maquina
8 @ IN SOA sd-ns1.2asir.com. admin.2asir.com. (
9     1 ; Serial - numero de serie de la zona
10    604800 ; Refresh - tiempo de refresco para los servers esclavos
11    86400 ; Retry - tiempo de reintento de contacto del esclavo al DNS
12    ;maestro
13    2419200 ; Expire - tiempo de expiracion de los servers esclavos
14    ;tiempo tras el cual el esclavo deja de contactar con el maestro
15    604800 ) ; Negative Cache TTL - tiempo que se almacena una consulta
16    ;no resuelta en cache, transcurrido se reintenta
17 ;Definimos nuestros servidores DNS autoritarios que resolveran consultas
18 @ IN NS sd-ns1.2asir.com.
19 @ IN NS sd-ns2.2asir.com.
20 ;El @ hace referencia al nombre del dominio(lo establecido en origen)
21 ;Establecemos la resolucion de dichos DNS, para que si se busca por nombre se resuelva
22 ;a IP.
23 ;indicamos el ultimo numero de la ip que queremos que apunte a un equipo
24 ;es el ultimo ya que la zona de busqueda inversa toma los primeros 24 bits de la ip
25 11 IN PTR sd-ns1.
26 12 IN PTR sd-ns2.
27 ;Los punteros de nuestros dos servidores
28
29 20 IN PTR www.
30 ;Puntero de la IP 172.16.30.20 a www
31 21 IN PTR www2.
32 ;Puntero de la 172.16.30.21 a www2
33 22 IN PTR intranet.
34 30 IN PTR mail.
35 ;ultimos dos punteros para la 172.16.30.22 y la 172.16.30.30
```

- 1 Pantallazo con el resultado de los dos chequeos de la zona creada.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:/etc/bind$ named-checkconf
administrador@sd-ns1:/etc/bind$ named-checkzone "30.16.172.in-addr.arpa" "/etc/bind/zones/db.30.16.1
72.in-addr.arpa"
zone 30.16.172.in-addr.arpa/IN: loaded serial 1
OK
administrador@sd-ns1:/etc/bind$
```

- 1 Pantallazo con la verificación del estado del servicio BIND en el servidor -ns1.2asir.com. No debe tener ningún error.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:/etc/bind$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2024-11-19 18:34:07 UTC; 47min ago
     Docs: man:named(8)
  Process: 2261 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
    Main PID: 2262 (named)
      Tasks: 14 (limit: 4520)
     Memory: 9.0M
        CPU: 181ms
    CGroup: /system.slice/named.service
            └─2262 /usr/sbin/named -u bind

nov 19 19:10:58 sd-ns1.2asir.com named[2262]: network unreachable resolving 'ubuntu.com/DS/IN': 200>
nov 19 19:10:58 sd-ns1.2asir.com named[2262]: network unreachable resolving 'ubuntu.com/DS/IN': 200>
nov 19 19:10:58 sd-ns1.2asir.com named[2262]: success resolving '_http._tcp.security.ubuntu.com/SRV'
nov 19 19:10:58 sd-ns1.2asir.com named[2262]: network unreachable resolving 'security.ubuntu.com/A'
nov 19 19:10:58 sd-ns1.2asir.com named[2262]: success resolving '_https._tcp.esm.ubuntu.com/SRV' af
nov 19 19:17:07 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001>
nov 19 19:17:07 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001>
nov 19 19:17:07 sd-ns1.2asir.com named[2262]: network unreachable resolving '2asir.com/NS/IN': 2001>
nov 19 19:17:08 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns1.2asir.com/A' after disablin
nov 19 19:17:08 sd-ns1.2asir.com named[2262]: success resolving 'sd-ns1.2asir.com/AAAA' after disab
```

Paso 4: Habilitar la transferencia de zona

Resultado de este Paso:

- 1 Pantallazo con el contenido de los archivos de declaración de zona de cada uno de los servidores.

```
SD_NS1 x SD_NS2 x
1 //
2 // Do any local configuration here
3 //
4
5 // Consider adding the 1918 zones here, if they are not used in your
6 // organization
7 //include "/etc/bind/zones.rfc1918";
8
9 zone "2asir.com" {
10     type master;
11     file "/etc/bind/zones/db.2asir.com";
12     allow-transfer { 172.16.30.12; };
13 };
14
15 zone "30.16.172.in-addr.arpa" {
16     type master;
17     file "/etc/bind/zones/db.30.16.172.in-addr.arpa";
18     allow-transfer { 172.16.30.12; };
19 };
```

```
SD_NS1 x SD_NS2 x
1 //
2 // Do any local configuration here
3 //
4
5 // Consider adding the 1918 zones here, if they are not used in your
6 // organization
7 //include "/etc/bind/zones.rfc1918";
8
9 zone "2asir.com" {
10     type slave;
11     file "/etc/bind/zones/db.2asir.com";
12     masters { 172.16.30.11; };
13 };
14
15 zone "30.16.172.in-addr.arpa" {
16     type slave;
17     file "/etc/bind/zones/db.30.16.172.in-addr.arpa";
18     masters { 172.16.30.11; };
19 };
```

- 2 Pantallazos con el resultado de los archivos de zona transferidos donde se vea claramente la ruta de cada uno de esos archivos.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind/zones$ cat db.2asir.com
;
2asircom7sd-ns12asircomadmin2asircom :Q$ :% :
2asircom
G :
2asircomsd-ns12asircom
comsd-ns22asircom3 :
2asircom
mail2asircom. :intranet2asircom* :mail2asircom, :sd-ns12asircom, :sd-ns22asircom
) :www2asircom* :www22asircom* :www32asircointranet2asircomadministrador@sd-ns2:/etc/bind/zones$
```

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind/zones$ cat db.30.16.172.in-addr.arpa
;3016172in-addrarpa7sd-ns12asircomadmin2asircom :Q$ :T :3016172in-addrarpa
d-ns12asircomsd-ns22asircom9 :
:113016172in-addrarpsd-ns19 :
:123016172in-addrarpsd-ns26 :
016172in-addrarpawww7 :
:213016172in-addrarpawww2; :
:223016172in-addrarpa
ntranet7 :
:303016172in-addrarpamailadministrador@sd-ns2:/etc/bind/zones$
```

- 2 pantallazos, uno por cada servidor donde se vea el estado del servicio BIND con el nombre de la máquina virtual visible. No debe tener ningún error.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns1:/etc/bind$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
  Active: active (running) since Tue 2024-11-19 20:11:50 UTC; 2min 8s ago
    Docs: man:named(8)
  Process: 1537 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
 Main PID: 1538 (named)
    Tasks: 14 (limit: 4520)
   Memory: 6.6M
      CPU: 58ms
   CGroup: /system.slice/named.service
           └─1538 /usr/sbin/named -u bind

nov 19 20:11:50 sd-ns1.2asir.com named[1538]: network unreachable resolving './NS/IN': 2001:500:1::
nov 19 20:11:50 sd-ns1.2asir.com named[1538]: network unreachable resolving './NS/IN': 2001:503:c27:
nov 19 20:11:50 sd-ns1.2asir.com named[1538]: network unreachable resolving './NS/IN': 2001:500:2::
nov 19 20:11:50 sd-ns1.2asir.com named[1538]: network unreachable resolving './NS/IN': 2001:dc3::35
nov 19 20:11:50 sd-ns1.2asir.com named[1538]: managed-keys-zone: Key 20326 for zone . is now truste
nov 19 20:11:50 sd-ns1.2asir.com named[1538]: resolver priming query complete: success
nov 19 20:12:07 sd-ns1.2asir.com named[1538]: client @0x7f951c0145c8 172.16.30.12#40541 (2asir.com)
nov 19 20:12:07 sd-ns1.2asir.com named[1538]: client @0x7f951c0145c8 172.16.30.12#40541 (2asir.com)
nov 19 20:12:07 sd-ns1.2asir.com named[1538]: client @0x7f952800d478 172.16.30.12#42169 (30.16.172.
nov 19 20:12:07 sd-ns1.2asir.com named[1538]: client @0x7f952800d478 172.16.30.12#42169 (30.16.172.
Lines 1-22/22 (END)
```

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind$ sudo systemctl status bind9.service
• named.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
  Active: active (running) since Tue 2024-11-19 20:12:07 UTC; 2min 6s ago
    Docs: man:named(8)
  Process: 1490 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
 Main PID: 1491 (named)
    Tasks: 14 (limit: 4520)
   Memory: 6.5M
      CPU: 49ms
   CGroup: /system.slice/named.service
           └─1491 /usr/sbin/named -u bind

nov 19 20:12:07 sd-ns2.2asir.com named[1491]: zone 2asir.com/IN: sending notifies (serial 1)
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: zone 30.16.172.in-addr.arpa/IN: Transfer started.
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: transfer of '30.16.172.in-addr.arpa/IN' from 172.16
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: zone 30.16.172.in-addr.arpa/IN: transferred serial
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: transfer of '30.16.172.in-addr.arpa/IN' from 172.16
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: transfer of '30.16.172.in-addr.arpa/IN' from 172.16
nov 19 20:12:07 sd-ns2.2asir.com named[1491]: zone 30.16.172.in-addr.arpa/IN: sending notifies (s
nov 19 20:12:08 sd-ns2.2asir.com named[1491]: timed out resolving './DNSKEY/IN': 1.1.1.1#53
nov 19 20:12:08 sd-ns2.2asir.com named[1491]: managed-keys-zone: Key 20326 for zone . is now trust
nov 19 20:12:13 sd-ns2.2asir.com named[1491]: resolver priming query complete: success
Lines 1-22/22 (END)
```

Paso 5: La utilidad dig

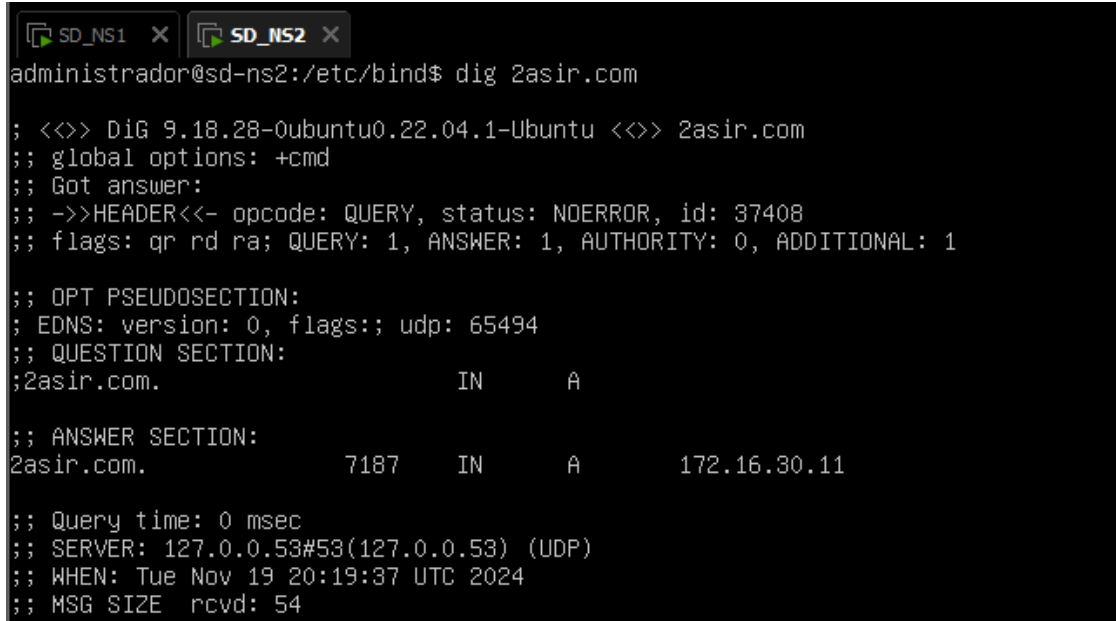
Con el comando dig, y solo con la ayuda que ofrece el comando, ejecuta lo siguiente (debes poner un pantallazo por cada ejecución):

- ¿Para qué sirve este comando?

El comando bind se utiliza para consultar servidores de nombres DNS

- Ejecuciones:

- Consulta para obtener la dirección IP de **www.2asir.com**



```
administrador@sd-ns2:/etc/bind$ dig 2asir.com

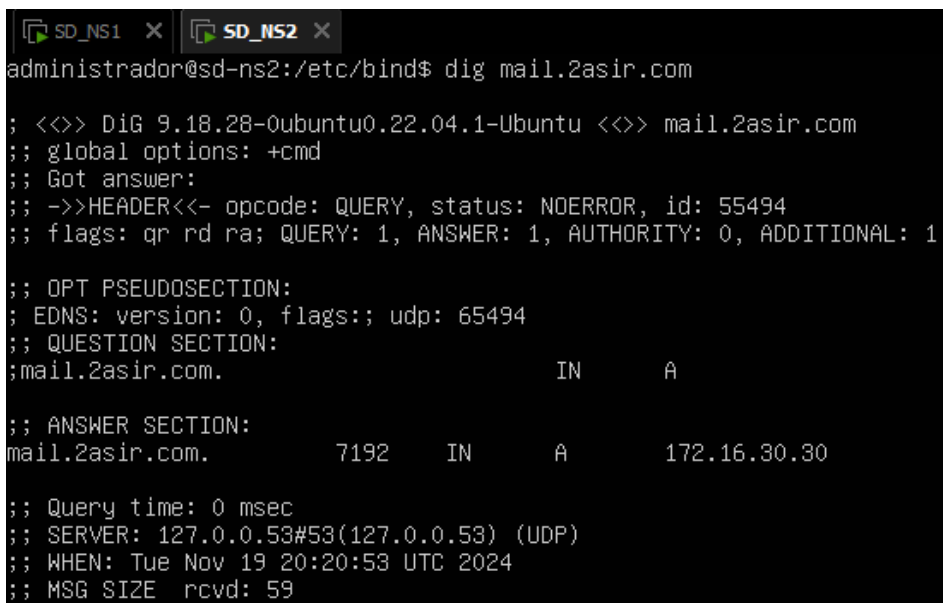
; <<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> 2asir.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37408
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;2asir.com.                IN      A

;; ANSWER SECTION:
2asir.com.                7187    IN      A      172.16.30.11

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Nov 19 20:19:37 UTC 2024
;; MSG SIZE rcvd: 54
```

- Consulta para obtener el servidor de correo para **2asir.com**



```
administrador@sd-ns2:/etc/bind$ dig mail.2asir.com

; <<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> mail.2asir.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55494
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;mail.2asir.com.          IN      A

;; ANSWER SECTION:
mail.2asir.com.          7192    IN      A      172.16.30.30

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Nov 19 20:20:53 UTC 2024
;; MSG SIZE rcvd: 59
```

- Consulta para obtener los servidores de nombres para 2asir.com

dig sd-ns1.2asir.com NS

- Consulta de búsqueda inversa para la IP 172.16.30.11

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind$ dig -x 172.16.30.11

; <<> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<> -x 172.16.30.11
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 40020
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;11.30.16.172.in-addr.arpa.      IN      PTR

;; ANSWER SECTION:
11.30.16.172.in-addr.arpa. 604800 IN      PTR      sd-ns1.

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Nov 19 20:26:25 UTC 2024
;; MSG SIZE rcvd: 74
```

- Consulta para obtener el alias de www3.2asir.com

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind$ dig www3.2asir.com

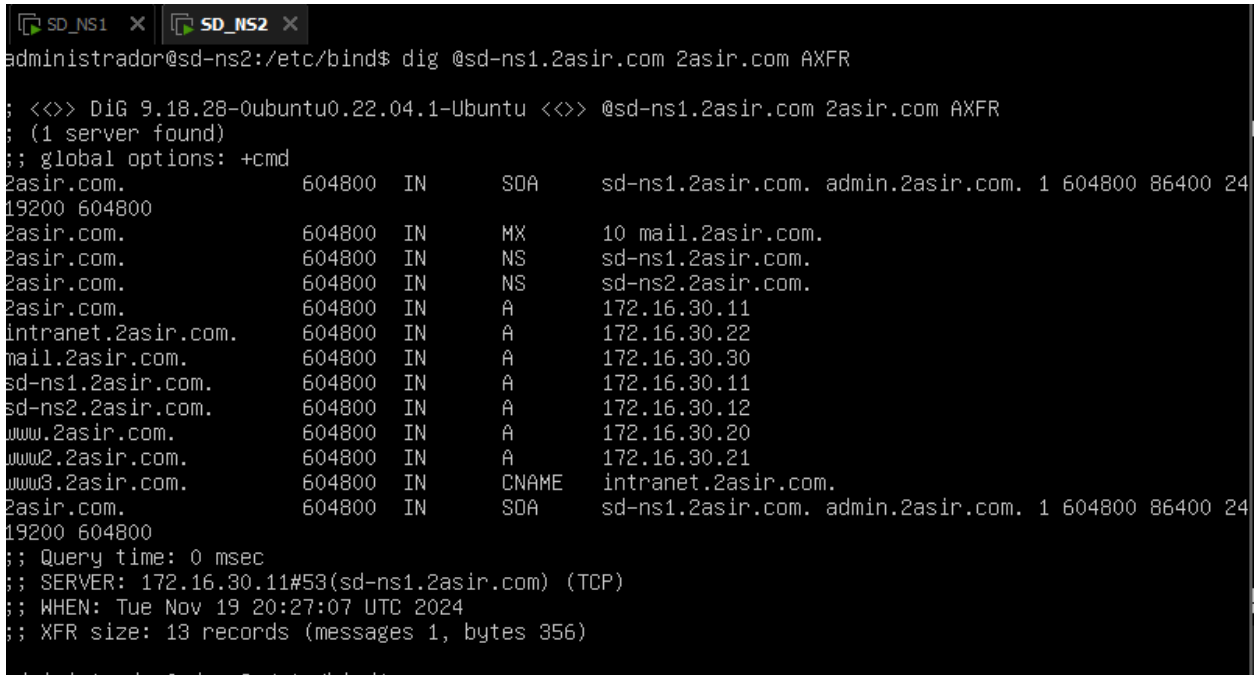
; <<> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<> www3.2asir.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43177
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www3.2asir.com.                IN      A

;; ANSWER SECTION:
www3.2asir.com.                604800 IN      CNAME    intranet.2asir.com.
intranet.2asir.com.            604800 IN      A        172.16.30.22

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Nov 19 20:22:32 UTC 2024
;; MSG SIZE rcvd: 82
```

- Ejecuta este comando: ■ **dig @ns1.2asir.com 2asir.com AXFR**



```
administrador@sd-ns2:/etc/bind$ dig @ns1.2asir.com 2asir.com AXFR

; <<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> @ns1.2asir.com 2asir.com AXFR
; (1 server found)
;; global options: +cmd
2asir.com.                604800 IN      SOA      sd-ns1.2asir.com. admin.2asir.com. 1 604800 86400 24
19200 604800
2asir.com.                604800 IN      MX       10 mail.2asir.com.
2asir.com.                604800 IN      NS       sd-ns1.2asir.com.
2asir.com.                604800 IN      NS       sd-ns2.2asir.com.
2asir.com.                604800 IN      A        172.16.30.11
intranet.2asir.com.       604800 IN      A        172.16.30.22
mail.2asir.com.           604800 IN      A        172.16.30.30
sd-ns1.2asir.com.         604800 IN      A        172.16.30.11
sd-ns2.2asir.com.         604800 IN      A        172.16.30.12
www.2asir.com.            604800 IN      A        172.16.30.20
www2.2asir.com.           604800 IN      A        172.16.30.21
www3.2asir.com.           604800 IN      CNAME    intranet.2asir.com.
2asir.com.                604800 IN      SOA      sd-ns1.2asir.com. admin.2asir.com. 1 604800 86400 24
19200 604800
;; Query time: 0 msec
;; SERVER: 172.16.30.11#53(sd-ns1.2asir.com) (TCP)
;; WHEN: Tue Nov 19 20:27:07 UTC 2024
;; XFR size: 13 records (messages 1, bytes 356)
```

- ¿Qué hace? Explícalo con el resultado.

Con @ le indicamos el servidor DNS a usar, posteriormente el dominio

Consultar IP de Google.

```
SD_NS1 x SD_NS2 x
administrador@sd-ns2:/etc/bind$ dig google.com

<<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> google.com
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8504
; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
; QUESTION SECTION:
google.com.                IN      A

; ANSWER SECTION:
google.com.                3       IN      A      142.250.184.14

; Query time: 0 msec
; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
; WHEN: Tue Nov 19 20:25:19 UTC 2024
; MSG SIZE  rcvd: 55
```